

MAT 535 HW09

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Problem 1 (§14.6#9). Determine the Galois group of $x^4 + 4x - 1$.

Solution. We can check that $p(x) = x^4 + 4x - 1$ is irreducible because it has no rational roots by the rational root theorem (± 1 are both clearly not roots) and it has no quadratic factors because if

$$p(x) = (x^2 + ax + c)(x^2 + bx + d) = x^4 + (a + b)x^3 + (ab + c + d)x^2 + (ad + bc)x + cd$$

then $cd = 1$, so $\{c, d\} = \{1, -1\}$; $a + b = 0$, so $a = -b$; $ab + c + d = -a^2 = 0$, so $a = 0$; and $ad + bc = 4$ and 0 simultaneously, contradiction.

The derivative $p'(x) = 4x^3 + 4$ has one real root, meaning $p(x)$ as a function $\mathbb{R} \rightarrow \mathbb{R}$ has one global minimum at $x = -1$, where $p(-1) = -2$. Thus, $p(x)$ has two real roots and two non-real roots. This means that the complex conjugation automorphism is a transposition of the two non-real roots.

The resolvent cubic of $p(x)$ is

$$q(x) = x^3 + 4x - 16 = (x - 2)(x^2 + 2x + 8).$$

This means the Galois group of $p(x)$ is either $\mathbb{Z}/(4)$ or D_8 by the classification in the text, but the former is impossible because cyclic subgroups of S_4 with order 4 do not have any transpositions, so the Galois group of $p(x)$ is dihedral with order 8. $\square$

Problem 2 (§14.6#19). Let $f(x)$ be an irreducible polynomial of degree 4 in $\mathbb{Q}[x]$ with discriminant D . Let K denote the splitting field of $f(x)$, viewed as a subfield of $\mathbb{C}$.

- (a) Prove that $\mathbb{Q}[\sqrt{D}] \subseteq K$.
- (b) Let τ denote complex conjugation and let τ_K denote the restriction of complex conjugation to K . Prove that τ_K is an element of $\text{Gal}(K/\mathbb{Q})$ with order 1 or 2 depending on whether every element of K is real or not.
- (c) Prove that if $D < 0$ then $\text{Gal}(K/\mathbb{Q})$ cannot be cyclic of order 4.
- (d) Prove generally that $\mathbb{Q}(\sqrt{D})$ for squarefree $D < 0$ is not a subfield of a cyclic quartic field.

Solution. (a) Let $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ be the roots of f . By definition,

$$\sqrt{D} = (\alpha_1 - \alpha_2)(\alpha_1 - \alpha_3)(\alpha_1 - \alpha_4)(\alpha_2 - \alpha_3)(\alpha_2 - \alpha_4)(\alpha_3 - \alpha_4)$$

which is a product of differences of elements of K , hence is in K , so $\mathbb{Q}[\sqrt{D}] \subseteq K$.

(b) Real numbers are fixed under complex conjugation, so if every element of K is real then τ_K fixes every element of K , hence has order 1. If K has any non-real elements then τ_K is not of order 1, but its order is at most 2 because τ has order 2 as an automorphism of $\mathbb{C}$, so τ_K must have order 2.

(c) If $\text{Gal}(K/\mathbb{Q})$ has order 4, then $K/\mathbb{Q}[\sqrt{D}]$ and K/K_+ are distinct degree 2 subextensions of $K/\mathbb{Q}$, where $K_+ = \mathbb{R} \cap K$ (they are distinct because $K_+ \subseteq \mathbb{R}$ and $\mathbb{Q}[\sqrt{D}]$ is not a subset of $\mathbb{R}$), which means $\text{Gal}(K/\mathbb{Q})$ has more than one element of order 2, so it is not cyclic.

(d) Again, if $K/\mathbb{Q}$ has degree 4, then $\mathbb{Q}[\sqrt{D}]$ and K_+ are distinct subfields that both correspond to order 2 elements of $\text{Gal}(K/\mathbb{Q})$, so the Galois group is not cyclic because it has too many elements of order 2. $\square$

Problem 3 (§14.6#28). Let α be a root of the irreducible polynomial $f(x) \in F[x]$ and let $K = F[\alpha]$. Let D be the discriminant of $f(x)$. Prove that $D = (-1)^{n(n-1)/2} N_{K/F}(f'(\alpha))$.

Solution. If α is not a separable element of F , then α has a multiplicity higher than 1 as a root of f , so $D = 0$ and $f'(\alpha) = 0$ implies that $N_{K/F}(f'(\alpha))$, as desired.

Assume now that α , hence $f(x)$, is separable, and let $\alpha = \alpha_1, \dots, \alpha_n$ be the roots of $f(x)$ in some splitting extension L/K of $f(x)$. Then

$$D = \prod_{i < j} (\alpha_i - \alpha_j)^2 = (-1)^{n(n-1)/2} \prod_{i \neq j} (\alpha_i - \alpha_j) = (-1)^{n(n-1)/2} \prod_{i=1}^n \prod_{j \neq i} (\alpha_i - \alpha_j),$$

where the coefficient of $(-1)^{n(n-1)/2}$ comes from negating $1 + 2 + \dots + n - 1$ (i.e. half of the) factors in the leftmost product where $i < j$. Then, $f'(x) = \sum_{i=1}^n \frac{f(x)}{x - \alpha_i}$, so every addend but the i th has a factor of $x - \alpha_i$, meaning $f'(\alpha_i) = \prod_{j \neq i} (\alpha_i - \alpha_j)$, so

$$D = (-1)^{n(n-1)/2} \prod_{i=1}^n f'(\alpha_i).$$

On the other hand, since $\text{Gal}(L/K)$ and $\text{Gal}(L/F)$ act transitively on $\{\alpha_1, \dots, \alpha_n\}$,

$$N_{K/F}(f'(\alpha))^{[L:K]} = N_{L/F}(f'(\alpha)) = \prod_{\sigma \in \text{Gal}(L/F)} \sigma(f'(\alpha)) = \left(\prod_{i=1}^n f'(\alpha_i) \right)^{[L:K]}$$

where the first equality is true because $N_{K/F}(f'(\alpha)) \in K$, so $D = (-1)^{n(n-1)/2} N_{K/F}(f'(\alpha))$. $\square$

Problem 4 (§14.7#2). Let ζ be a primitive 7th root of unity and let $\alpha = \zeta + \zeta^{-1}$.

- (a) Show that ζ is a root of $z^2 - \alpha z + 1 \in \mathbb{Q}(\alpha)[z]$.
- (b) Show using the minimal polynomial for ζ that α is a root of the cubic $x^3 + x^2 - 2x - 1$.
- (c) Use (a) and (b) together with the explicit solution of the cubic in (b) to express ζ in terms of radicals.

Solution. (a) Plugging in $z = \alpha$, we compute

$$\zeta^2 - (\zeta + \zeta^{-1})\zeta + 1 = \zeta^2 - \zeta^2 - 1 + 1 = 0.$$

(b) Since $\zeta^6 + \zeta^5 + \zeta^4 + \zeta^3 + \zeta^2 + \zeta + 1 = 0$, dividing by ζ^3 gives

$$\alpha^3 - \alpha^2 - 2\alpha - 1 = (\alpha^3 - 3\alpha) + (\alpha^2 - 2) + \alpha + 1 = (\zeta^3 + \zeta^{-3}) + (\zeta^2 + \zeta^{-2}) + (\zeta + \zeta^{-1}) + 1 = 0.$$

(c) no thanks (The idea is of course to pick a root of the cubic to be α and use the fact that α is twice the real part of ζ and $|\zeta| = 1$ to compute the imaginary part of ζ , but I am not writing that out.) $\square$

Problem 5 (§14.7#7). Let F be a field of characteristic not dividing n containing the n th roots of unity and let K be a cyclic extension of degree d dividing n . Then $K = F[\sqrt[n]{a}]$ for some nonzero $a \in F$. Let σ be a generator for $\text{Gal}(K/F)$.

- (a) Show that $\sigma(\sqrt[n]{a}) = \zeta \sqrt[n]{a}$ for some primitive d th root of unity ζ .
- (b) Suppose $K = F[\sqrt[n]{a}] = F[\sqrt[n]{b}]$. Use (a) to show that

$$\frac{\sigma(\sqrt[n]{a})}{\sqrt[n]{a}} = \left(\frac{\sigma(\sqrt[n]{b})}{\sqrt[n]{b}} \right)^i$$

for some integer i relatively prime to d . Conclude that σ fixes $\frac{\sqrt[n]{a}}{(\sqrt[n]{b})^i}$, so this is an element of F .

- (c) Prove that $K = F[\sqrt[n]{a}] = F[\sqrt[n]{b}]$ if and only if $a = b^i c^n$ and $b = a^j d^n$ for some $c, d \in F$, i.e. a and b represent the same subgroup of $F^\times / (F^\times)^n$.

Solution. (a) The minimal polynomial of $\sqrt[n]{a}$ divides $x^n - a$, so the Galois conjugates of $\sqrt[n]{a}$ are $\sqrt[n]{a}$ times some n th root of unity, so $\sigma(\sqrt[n]{a}) = \zeta \sqrt[n]{a}$ for an n th root of unity ζ . The order of ζ in $F^\times$ is equal to the order of σ in $\text{Gal}(K/F)$ because σ is determined by its action on $\sqrt[n]{a}$. This order is equal to d , so ζ is a primitive d th root of unity.

(b) By (a), $\sigma(\sqrt[n]{a}) = \zeta_1 \sqrt[n]{a}$ and $\sigma(\sqrt[n]{b}) = \zeta_2 \sqrt[n]{b}$ for primitive d th roots of unity ζ_1 and ζ_2 . These roots of unity both generate the cyclic group of roots of unity μ_d , so some power ζ_2^i of ζ_2 must equal ζ_1 as desired. Rearranging the resulting equality then shows

$$\sigma \left(\frac{\sqrt[n]{a}}{(\sqrt[n]{b})^i} \right) = \frac{\sigma(\sqrt[n]{a})}{(\sigma(\sqrt[n]{b}))^i} = \frac{\sqrt[n]{a}}{(\sqrt[n]{b})^i},$$

which means that $\frac{\sqrt[n]{a}}{(\sqrt[n]{b})^i}$ is fixed by σ , which only fixes F because it generates $\text{Gal}(K/F)$.

(c) If $a = b^i c^n$ and $b = a^j d^n$ then $\sqrt[n]{a} = c(\sqrt[n]{b})^i$ implies $\sqrt[n]{a} \in F[\sqrt[n]{b}]$ and similarly $\sqrt[n]{b} = d(\sqrt[n]{a})^j$ implies $\sqrt[n]{b} \in F[\sqrt[n]{a}]$. Conversely, if $\sqrt[n]{a}$ and $\sqrt[n]{b}$ generate the same extension of F , then (b) implies that $\sqrt[n]{a} = c(\sqrt[n]{b})^i$ for some $c \in F$, and applying (b) again after swapping a and b shows that $\sqrt[n]{b} = d(\sqrt[n]{a})^j$ for some integer j and $d \in F$. $\square$